Multi-Modal Sentiment Analysis

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Github: https://github.com/Wakamo-0716/ESE-527-Project.git

1.- Introduction -

1.1 Decisions to be impacted

This project is designed to support three key decisions in multimodal sentiment analysis. First, it informs model architecture selection, specifically whether sequential models such as LSTMs or more attention-based models are better suited for temporally aligned multimodal sentiment regression. Second, it supports fusion strategy selection, helping determine which fusion mechanism—Early Fusion, Gated Fusion, Cross-modal Attention, or Tensor Fusion—offers the best trade-off between prediction performance, robustness, and model complexity. Third, it informs the decision of whether multimodal modeling is worth the additional cost relative to unimodal baselines, especially in settings where collecting and processing audio and visual signals requires more resources than text alone.

1.2 Business value

On the business side, accurate multimodal sentiment recognition models can significantly enhance the intelligence of human-centered interactive systems. This has key applications in the following areas:

- **Customer Service and Opinion Monitoring**: By jointly analyzing speech, text, and facial expressions, systems can monitor emotional fluctuations in real time and detect potential negative experiences early, thereby improving customer retention and service quality.
- **Intelligent Recommendation and Marketing Optimization**: Emotion-aware models can increase the contextual relevance of advertisements and content recommendations, enabling more precise user profiling and adaptive marketing strategies.
- **Affective Computing and Enhanced Human-Computer Interaction**: In domains such as social robotics, virtual assistants, and educational technology, systems

capable of recognizing emotions can interact with users more naturally and empathetically, greatly improving user engagement and satisfaction.

1.3 Why do we care about this project?

We care about this project for both research and practical reasons. From a research perspective, multimodal learning is an important and active area in machine learning, and sentiment intensity prediction is a particularly challenging task because it requires modeling subtle emotional variation rather than only positive-versus-negative polarity. From a practical perspective, real human communication is inherently multimodal. As described in our first report, a speaker may produce neutral words while their tone and facial expression reveal a very different emotional state. This makes sentiment prediction an ideal problem for studying how textual, acoustic, and visual information complement each other. Our project is therefore motivated by a desire to understand not only which model performs best, but also why multimodal fusion helps, when it helps, and how much improvement it can provide over strong unimodal baselines.

1.4 Data assets

The primary data asset used in this project is the CMU-MOSEI dataset, a large-scale multimodal benchmark for sentiment and emotion analysis. CMU-MOSEI contains over 65 hours of annotated video from more than 1,000 speakers and over 23,000 sentence-level segments, with text, audio, and visual modalities temporally aligned at the utterance level. The dataset provides continuous sentiment labels ranging from -3 to +3, which makes it especially suitable for our regression-based formulation of sentiment intensity prediction.

2.- Data Preprocessing -

2.1 Raw Data Structure

The original dataset was provided in a pickle file with three predefined splits: training, validation, and test. Each split contains five fields: text, audio, vision, labels, and id. The three modalities are already aligned to a fixed sequence length of 50 time steps, with shapes $(N, 50, 300)$ for text, $(N, 50, 74)$ for audio, and $(N, 50, 35)$ for vision. The labels have shape $(N, 1, 1)$ and correspond to continuous sentiment scores, while id is used only for sample identification.

Since the modalities were already aligned, no additional feature extraction or synchronization was required. The preprocessing stage therefore focused on cleaning, anomaly analysis, and converting the data into a model-ready format.

2.2 Data Cleaning

Our preprocessing strategy was lightweight and focused on preserving the original temporal structure. We first verified that all three splits contained the required fields and that the numbers of samples were consistent across text, audio, vision, labels, and id. We also confirmed that all modalities shared the same aligned sequence length of 50 time steps.

We then cleaned the feature tensors by replacing any NaN or Inf values with zero, while keeping the labels unchanged. In addition, all modality features were converted from float64 to float32 to reduce memory usage and improve compatibility with deep learning frameworks, and the labels were reshaped from $(N, 1, 1)$ to $(N,)$ for regression training. Finally, because the three modalities have different scales and dimensionalities, we normalized text, audio, and vision separately using the mean and standard deviation computed only from the training split, and applied the same statistics to the validation and test splits to avoid data leakage.

2.3 Outlier Detection

2.3.1 Isolation Forest for Sample-level Outlier Detection

We applied Isolation Forest to detect atypical samples in the multimodal feature space. This method isolates anomalies through recursive random partitioning, where samples with shorter average path lengths are considered more anomalous. Using the concatenated multimodal representation and a contamination setting of 0.05, the model flagged about 5% of the training samples as outliers.

However, this ratio should not be interpreted as the true proportion of corrupted data. Instead, it represents the most atypical subset under the chosen setting.

2.3.2 Temporal Instability and Boundary Effects

We also examined the temporal mean and temporal standard deviation of each modality across the 50 aligned time steps. The results show that all three modalities become less stable in the later part of the sequence, with more noticeable changes after approximately time step 40.

This pattern is likely caused by boundary effects introduced when variable-length original sequences were aligned or truncated to a fixed length of 50. Among the three modalities, audio shows the strongest late-stage fluctuation, suggesting that it is more sensitive to local instability and alignment artifacts.

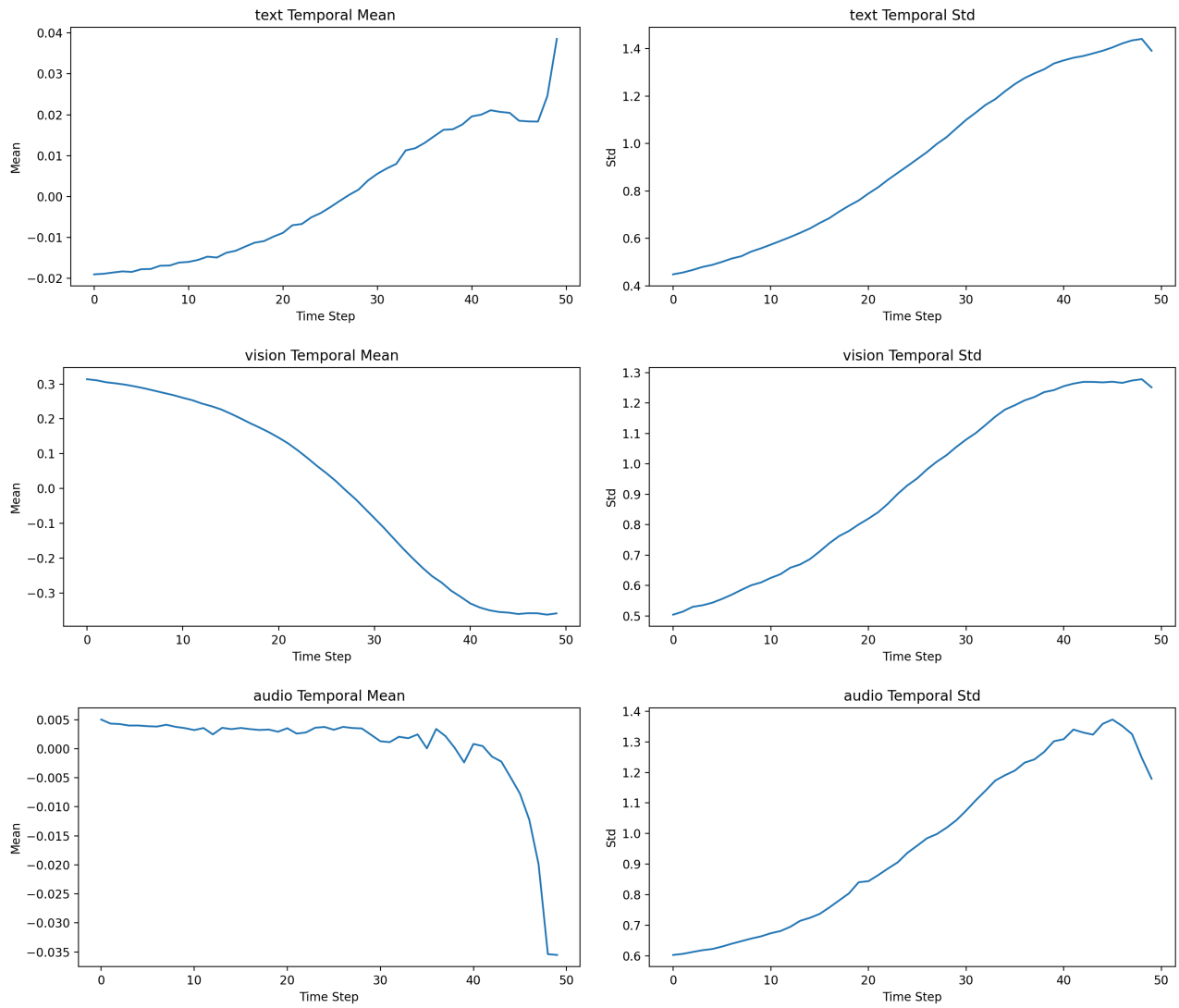


Figure 2.1 Temporal mean and std of text, vision, and audio modalities

2.3.3 Interpretation and Handling Strategy

To better understand the flagged outliers, we analyzed the relationship between Isolation Forest anomaly scores and several dataset characteristics. The correlation between anomaly score and label magnitude was very weak ($r \approx -0.047$), while the correlations with temporal jump intensity were much stronger: about -0.928 for text, -0.854 for audio, and -0.684 for vision.

These results suggest that the detected outliers are mainly associated with temporal instability rather than extreme sentiment labels. Therefore, we did not remove them from the dataset. Instead, we retained all samples and used the analysis to guide downstream model design, especially the use of global sequence aggregation rather than relying only on the final time step.

2.4 Data Encoding for Model Training

After preprocessing, the dataset was organized into a model-ready format:

- text: $(N, 50, 300)$
- audio: $(N, 50, 74)$
- vision: $(N, 50, 35)$
- labels: $(N,)$

This representation preserves the aligned sequence structure and can be directly used for both unimodal and multimodal experiments. No pooling, flattening, or fusion was performed during preprocessing, which keeps the data flexible for later comparisons across LSTM- and Transformer-based models.

2.5 Label Distribution and Task Implication

We also examined the distribution of sentiment labels. Most labels are concentrated around zero, while strongly positive and strongly negative samples are relatively rare. This indicates that the regression target space is imbalanced, with moderate sentiment values appearing much more frequently than extreme cases.

This property suggests that downstream evaluation should include multiple regression metrics, such as MAE, RMSE, and correlation, rather than relying on a single measure.

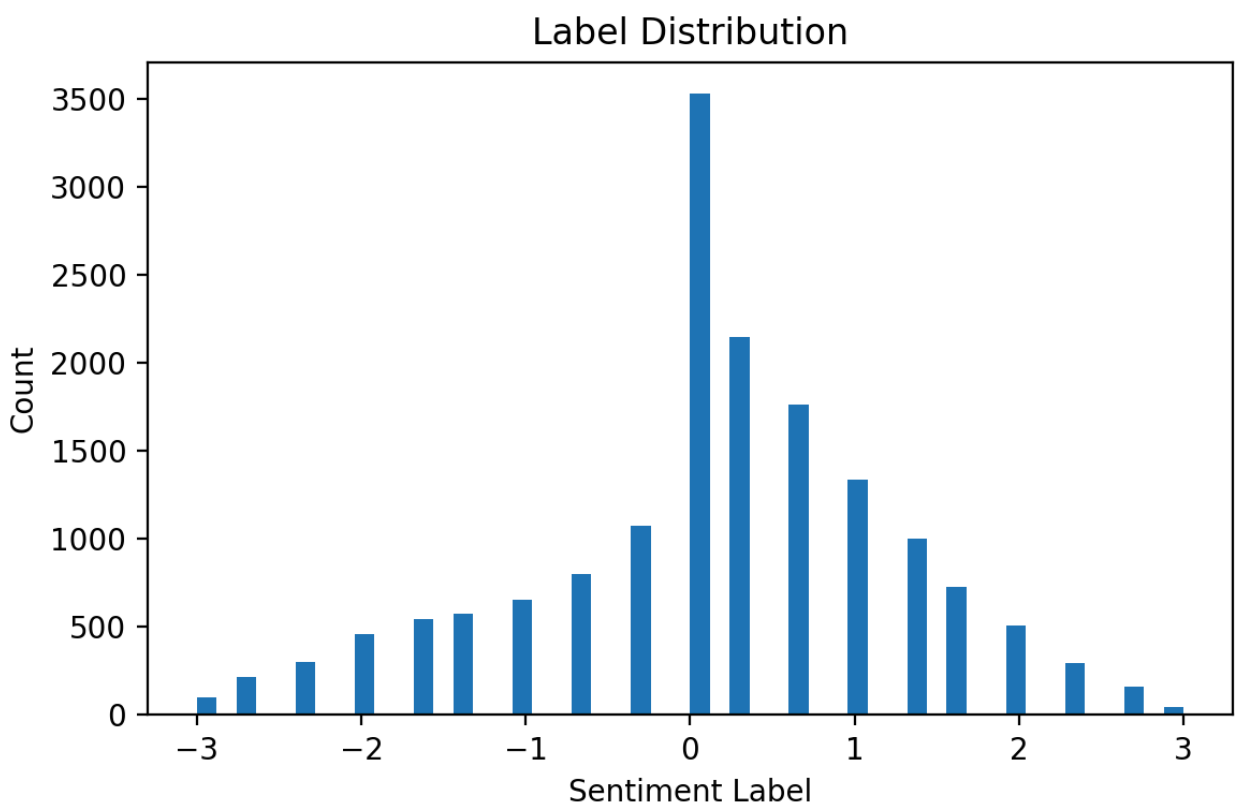


Figure 2.2 Label distribution of sentiment scores

2.6 Summary

In summary, the preprocessing pipeline preserved the original aligned sequence structure while performing essential cleaning, normalization, and anomaly analysis. The results show that the main source of irregularity in the dataset is temporal instability, especially near the later time steps and particularly in the audio modality. After preprocessing, the data are stored in a clean and model-ready sequential format suitable for both LSTM- and Transformer-based experiments.

3.- Model Updates -

3.1 Overall modeling strategy

Our project focuses on multimodal sentiment regression using temporally aligned text, audio, and vision features from the CMU-MOSEI dataset. To support this task, we study two major modeling dimensions simultaneously:

1. Sequence modeling architecture

- LSTM-based models
- Transformer-based models

2. Multimodal fusion strategy

- Early Fusion
- Gated Fusion
- Cross-modal Attention
- Tensor Fusion

Our approach first focuses on identifying the best-performing multimodal architecture, and then uses that model framework to compare multimodal and single-modality inputs in a more controlled way.

3.2 Sequence modeling architectures

To capture temporal dependencies in aligned multimodal sequences, we explored two sequence modeling architectures: LSTM and Transformer.

LSTM

LSTM is well suited for temporally aligned data because it processes sequential information step by step and preserves contextual information over time. In our implementation, the LSTM encoder produces hidden states for all time steps rather than relying only on the final hidden state. We then apply temporal mean pooling to obtain a global sequence representation. This helps preserve sentiment cues distributed across the full utterance.

Transformer

Transformer is also suitable for aligned multimodal sequences because it models temporal dependencies through self-attention, allowing each time step to directly attend to all other positions in the sequence. This is particularly useful when sentiment information is distributed unevenly across an utterance or when important cues appear far apart in time. In our implementation, each modality is first projected into a shared hidden space before being passed through the Transformer encoder. Positional encoding is added to preserve temporal order, since the attention mechanism itself does not encode sequence position.

3.3 Multimodal fusion strategies

We implemented four multimodal fusion strategies and combined them with our sequence modeling architectures.

(a) Early Fusion

In Early Fusion, the text, audio, and vision features are combined at the input level before sequence encoding. This serves as the most direct and simple multimodal baseline. The main advantage of this strategy is simplicity, but it does not explicitly model modality-specific importance or inter-modal relationships.

(b) Gated Fusion

In Gated Fusion, the model learns adaptive weights for the modality representations before combining them. This allows the model to assign different importance to each modality depending on the input sample. The motivation for this strategy is that different utterances may rely on different modalities for sentiment expression.

(c) Cross-modal Attention

In Cross-modal Attention, attention mechanisms are used to explicitly model dependencies across modalities. This allows one modality to focus on the most relevant information from another modality. In sentiment analysis, this is especially useful because text often provides semantic content, while audio and vision provide emotional context.

(d) Tensor Fusion

Tensor Fusion models higher-order interactions among modalities by explicitly constructing cross-modal combinations. This strategy is more expressive than simple concatenation, since it captures unimodal, bimodal, and trimodal relationships. However, it is also computationally heavier and more sensitive to dimensionality and regularization choices.

3.4 Model selection strategy

We compare the multimodal models created by combining our sequence backbones with the four fusion strategies. The goal of this stage is to determine which multimodal architecture provides the strongest sentiment regression performance.

After identifying the best-performing multimodal model, we use that same overall model framework as a reference architecture and train it on single modalities separately. In other words, we apply the strongest model design to:

- text-only input
- audio-only input
- vision-only input

and then compare those results against the full multimodal version.

3.5 Modality projection

One challenge in this project is that the three modalities have very different raw feature dimensions:

- Text: 300
- Audio: 74
- Vision: 35

This dimensional imbalance may cause the text modality to dominate the shared representation space during multimodal learning. To address this, we introduced modality-specific projection layers that map each modality into a shared latent dimension before fusion and sequence encoding.

3.6 Performance metrics

Because this is a regression task, we evaluate all models using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- Pearson Correlation

MAE and RMSE measure prediction error, where lower values indicate better performance. Pearson correlation measures the agreement between predicted sentiment scores and ground-truth labels, where higher values indicate better alignment.

3.7 Machine Learning Morphism (MLM) workflow

Our current machine learning workflow can be summarized as:

Aligned multimodal input features (text, audio, vision)

1. Data Cleaning and Preprocessing
2. Modality Projection and Normalization
3. Sequence Encoding (LSTM or Transformer)
4. Multimodal Fusion (Early Fusion, Gated Fusion, Cross-modal Attention, or Tensor Fusion)
5. Temporal Aggregation and Task-Specific Representation
6. Regression Head
7. Continuous Sentiment Prediction
8. Evaluation Using MAE, RMSE, and Pearson Correlation

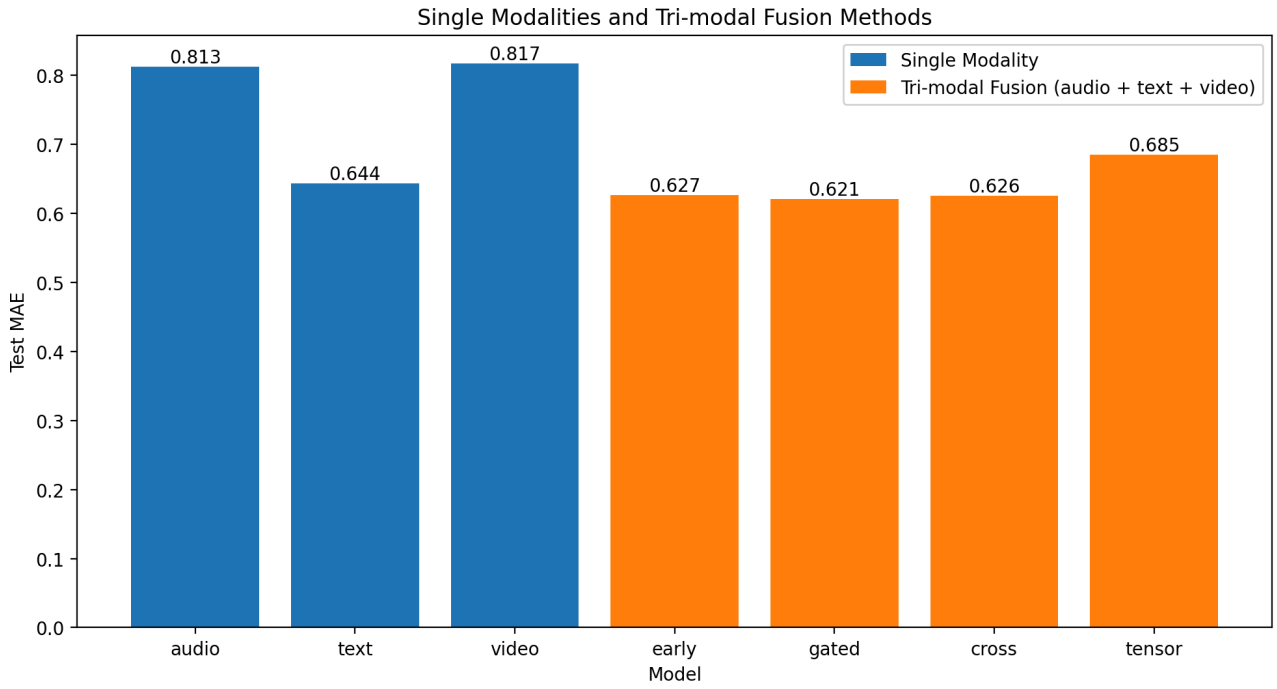
This workflow provides a unified framework for comparing multiple architectures and fusion strategies under the same multimodal sentiment regression task.

4.- Current Results -

4.1 Transformer

4.1.1 Preliminary Fusion Comparison

As an initial step, we conducted a preliminary experiment to compare the performance of four fusion strategies—Early Fusion, Gated Fusion, Cross-modal Fusion, and Tensor Fusion—in order to identify the most promising architecture for subsequent training. To make this comparison more informative, we also included the three unimodal settings, namely audio-only, text-only, and vision-only, as baselines. This design allows us to evaluate not only the relative effectiveness of different fusion methods, but also the extent to which multimodal fusion improves over single-modality learning.



As shown in the bar chart, the text-only model achieves the best performance among the three unimodal settings. Among the tri-modal fusion approaches, Gated Fusion, Cross-modal Fusion, and Early Fusion all demonstrate competitive performance, and each yields an improvement over the text-only baseline. Since this stage serves only as a coarse preliminary experiment, we selected Gated Fusion and Cross-modal Fusion for further hyperparameter tuning in the next stage, while retaining the text-only model and Early Fusion as baselines for comparison. Tensor Fusion was not pursued further, not only because its performance was inferior to the other three fusion methods, but also because its training time was substantially longer than that of the remaining approaches.

4.1.2 Hyperparameter Tuning

We performed a targeted hyperparameter search for the main multimodal fusion models, focusing primarily on Gated Fusion and Cross-modal Attention. Rather than using a fully exhaustive grid search, we explored a restricted set of influential parameters to balance model quality and computational cost.

For both models, the main search space included:

- Hidden dimension: 128, 256
- Dropout: 0.1, 0.2
- Learning rate: 1e-4, 3e-4

For Cross-modal Attention, we additionally tuned:

- Number of attention heads: 2, 4
- Number of Transformer layers: 1, 2

Other settings were kept fixed during tuning:

- Batch size: 32
- Weight decay: 1e-4
- Random seed: 42

The best configuration for each model was selected primarily based on validation MAE, while test MAE, RMSE, and correlation were used for final comparison and analysis. This helped ensure that model selection was guided by validation performance rather than direct optimization on the test set.

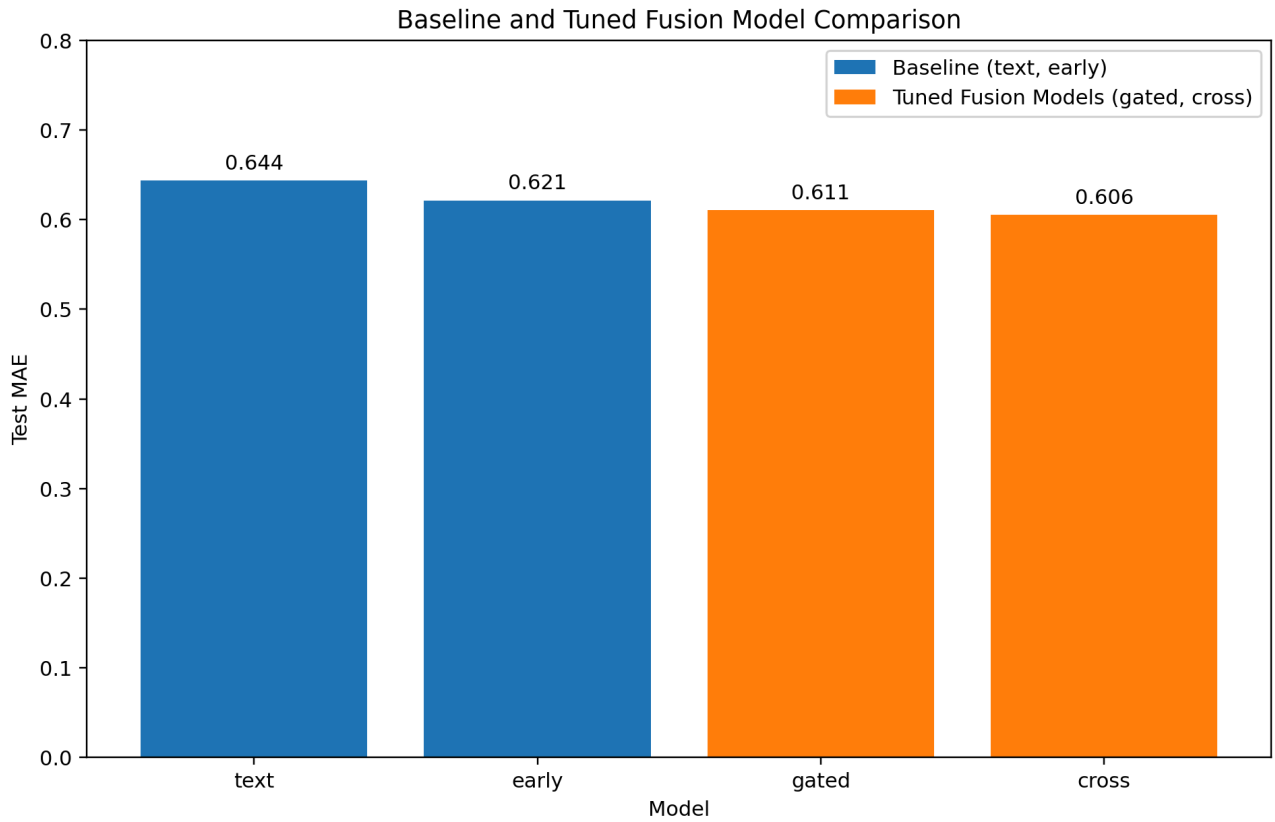


Table: Bootstrap Test Results

Comparison	95% CI Low	95% CI High	p-value	Significant	Better Model
early vs cross	0.0070	0.0246	0.0007	Yes	cross
early vs gated	0.0028	0.0189	0.0101	Yes	gated
gated vs cross	-0.0033	0.0132	0.2221	No	cross

After hyperparameter tuning, we selected the best-performing configurations of Gated Fusion and Cross-modal Fusion and compared them against two baselines, namely the

text-only model and Early Fusion. The results show that both tuned fusion models outperform the two baselines in terms of predictive performance. In particular, both Gated Fusion and Cross-modal Fusion achieve lower error than the text-only setting, indicating that multimodal fusion provides clear benefits over relying on textual information alone. They also improve upon Early Fusion, suggesting that more structured fusion mechanisms are more effective than simple early-stage feature concatenation.

These observations are further supported by the bootstrap significance tests. From a statistical inference perspective, the improvements of both Gated Fusion and Cross-modal Fusion over the baseline models are supported by the significance results, which strengthens the reliability of the empirical comparison. However, although Cross-modal Fusion achieves slightly better numerical performance than Gated Fusion, the statistical tests do not provide sufficient evidence to conclude that one of the two fusion methods is definitively superior to the other. Therefore, in the following stage, we retained the best hyperparameter settings for both models and continued with bi-modal fusion experiments.

4.1.3 Performance Comparison Across Unimodal, Bimodal, and Trimodal Fusion Settings

In this stage, we extended the comparison from unimodal models to bimodal and trimodal fusion models. All bimodal and trimodal experiments were trained using the best hyperparameter settings selected in the previous stage, and the results for Gated Fusion and Cross-modal Fusion are reported separately to facilitate a clearer comparison between the two fusion mechanisms.

Table: Cross Group Comparison Results

Comparison	95% CI Low	95% CI High	p-value	Significant
text+audio vs tri-modal	0.0028	0.0196	0.0075	Yes
text+vision vs tri-modal	-0.0005	0.016	0.0636	No
audio+vision vs tri-modal	0.1857	0.2202	0.0	Yes

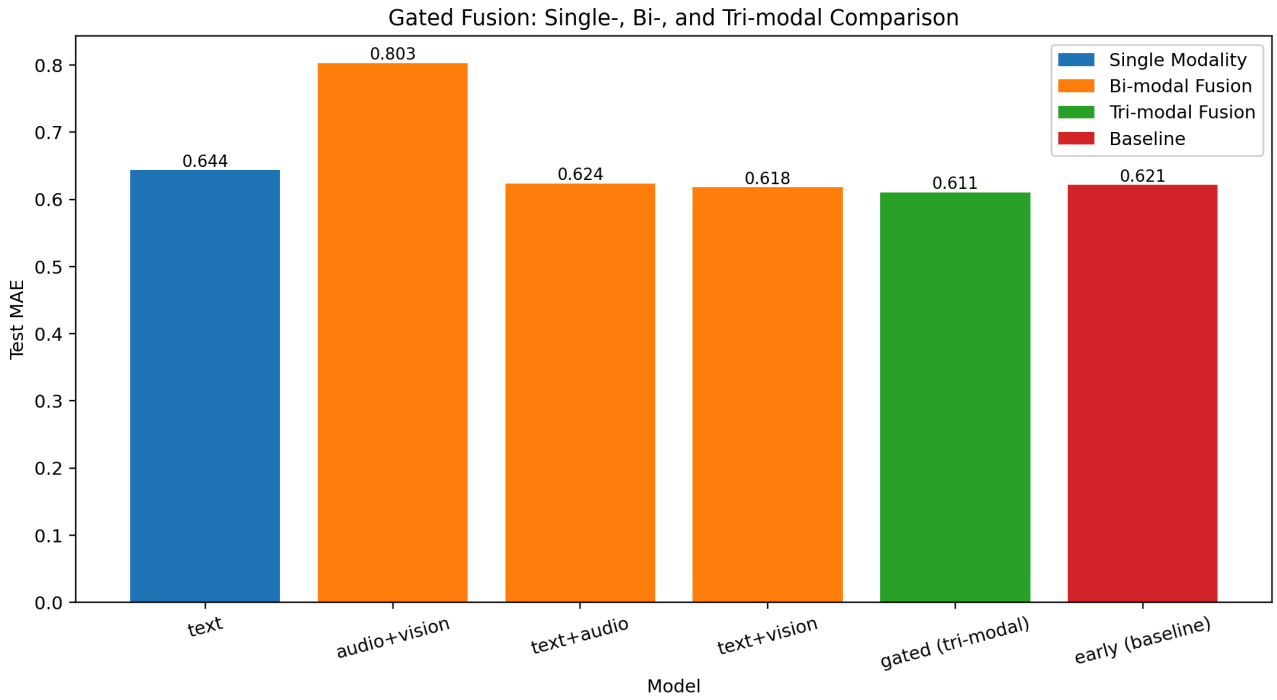


Table: Gated Group Comparison Results

Comparison	95% CI Low	95% CI High	p-value	Significant
text+audio vs tri-modal	0.0044	0.0215	0.0038	Yes
text+vision vs tri-modal	-0.0027	0.0171	0.153	No
audio+vision vs tri-modal	0.1765	0.2085	0.0	Yes

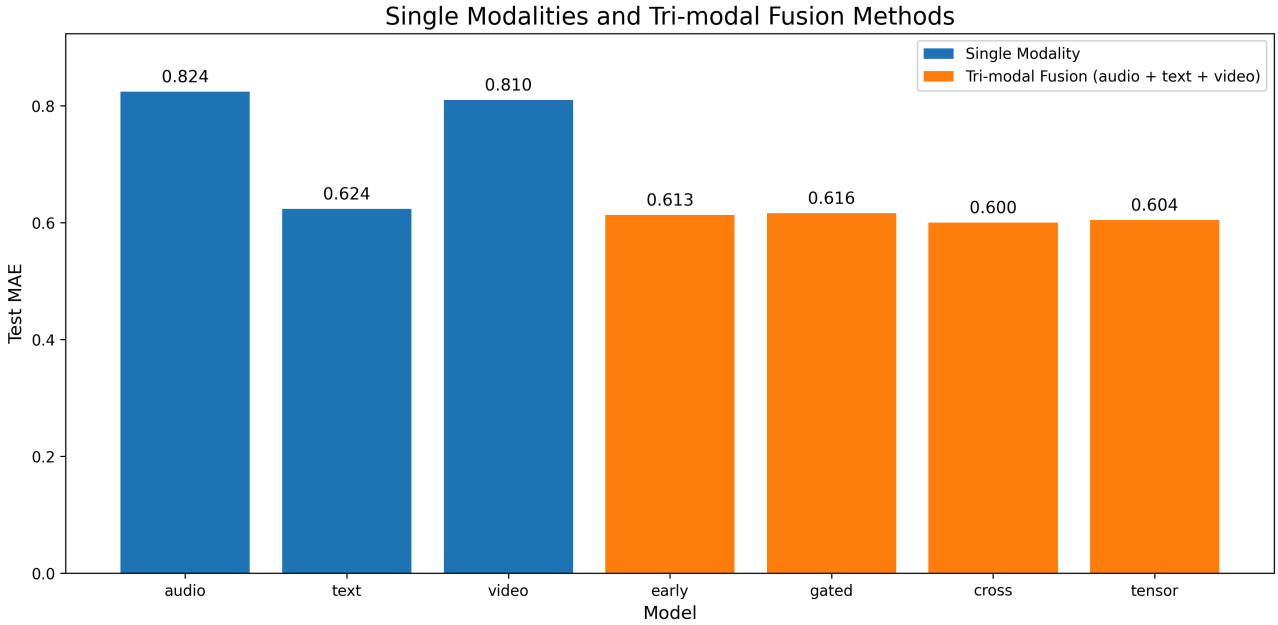
When comparing unimodal and bimodal settings, we observe that combining text with either audio or vision consistently improves performance relative to the text-only baseline. In contrast, the audio+vision combination performs worse than the text-only model, suggesting that text is the most informative modality in this task and that excluding it leads to a noticeable loss in predictive power. When the trimodal setting is further considered, both text+audio and text+vision yield higher MAE than the corresponding trimodal models. However, from the perspective of statistical inference, the difference between text+vision and the trimodal setting is not statistically significant. This pattern is consistent across both Gated Fusion and Cross-modal Fusion.

The bootstrap results help quantify whether the observed performance gaps are stable under resampling. In this context, a significant result indicates that the observed improvement is unlikely to be explained solely by sampling variability in the test set, whereas a non-significant result suggests that the current evidence is not strong enough to support a confident conclusion about superiority. Therefore, the non-significant difference between text+vision and the trimodal setting should not be interpreted as evidence that the two settings are equivalent, but rather as insufficient statistical evidence to confirm a reliable advantage under the current evaluation setting.

Overall, these findings suggest that the raw audio modality may be noisier or less informative than the other two modalities, whereas text provides the strongest predictive signal. At the same time, the best overall performance is still achieved by the trimodal models, indicating that incorporating all three modalities remains beneficial. In addition, Cross-modal Fusion consistently produces lower MAE than Gated Fusion across the corresponding settings, suggesting that Cross-modal Fusion is the more effective fusion strategy within the transformer-based framework considered in this work.

4.2 LSTM Results

4.2.1 Preliminary Fusion Comparison



4.2.2 Hyperparameter Tuning

We adopted a targeted hyperparameter tuning strategy instead of a fully exhaustive search. Based on our preliminary experiments, we identified **Cross-modal Attention** as the strongest multimodal model, and therefore focused tuning on this architecture. To ensure a fairer unimodal-versus-multimodal comparison, we also tuned the text-only model, since it was the strongest unimodal baseline.

For both models, we varied:

- Hidden dimension: 128, 256
- Number of layers: 1, 2
- Dropout: 0.1, 0.2, 0.3
- Learning rate: 5e-4, 1e-3

The following settings were fixed:

- Batch size: 32
- Weight decay: 1e-5
- Random seed: 42

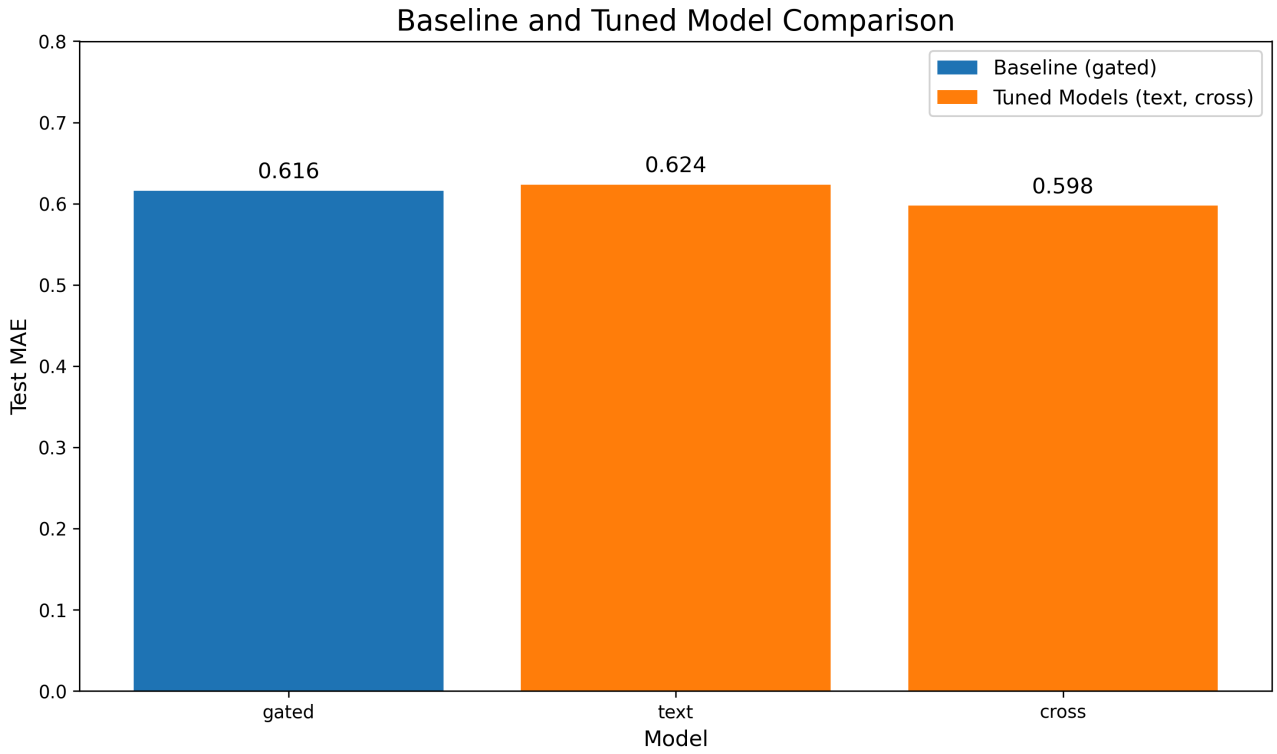


Table: Bootstrap Comparison Results

Comparison	95% CI Low	95% CI High	p-value	Significant	Better Model
text vs cross	0.0166	0.0353	< 0.0001	Yes	cross
gated vs cross	0.0110	0.0261	< 0.0001	Yes	cross
cross vs text	-0.0353	-0.0166	< 0.0001	Yes	cross

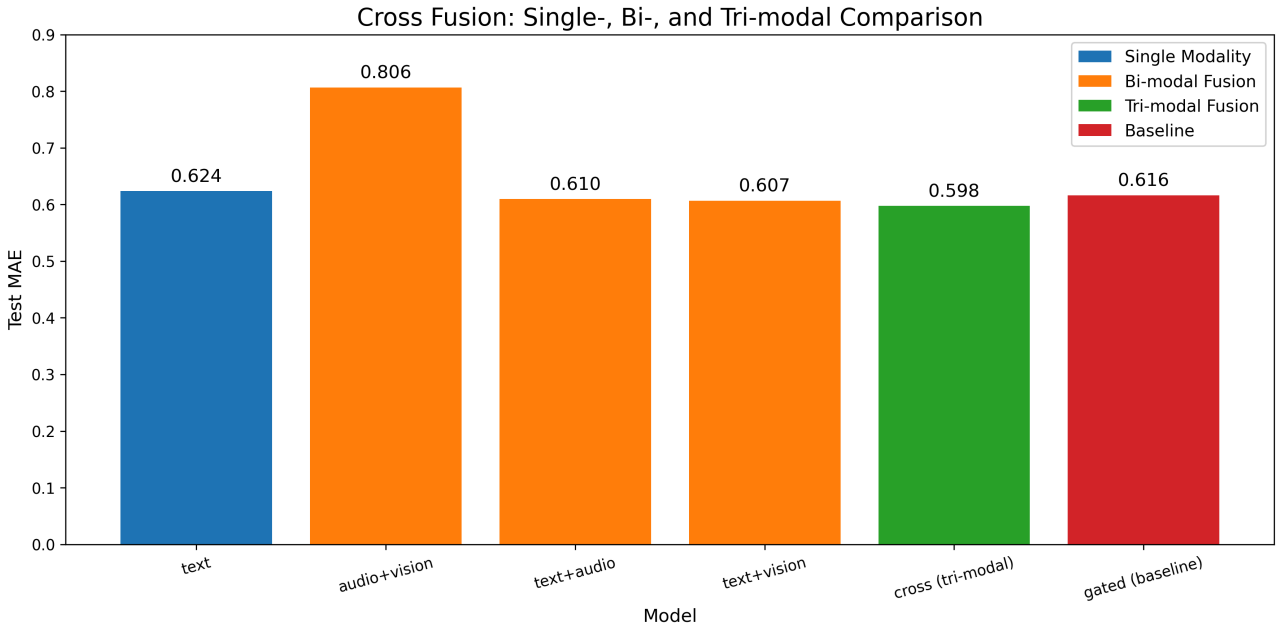
Following hyperparameter optimization, we retained the strongest configurations of the **text-only model** and the **Cross-modal Attention model**, and used **Gated Fusion** as an additional multimodal reference for comparison. The empirical results indicate that the tuned Cross-modal Attention model achieves the best overall predictive performance among the selected models, as reflected by its lower error metrics and higher correlation with the ground-truth sentiment scores. This pattern suggests that integrating multiple modalities remains advantageous even when the strongest unimodal baseline has been tuned, and further implies that explicitly modeling inter-modal dependencies is more effective than relying on a simpler gated fusion mechanism in this task.

The pairwise statistical comparisons further reinforce this conclusion. In both the text versus cross-modal attention and gated fusion versus cross-modal attention comparisons, the bootstrap confidence intervals exclude zero and the corresponding p-values indicate statistically reliable differences. Taken together, the results support two main conclusions: first, multimodal fusion contributes meaningful information beyond

text alone; second, among the models considered at this stage, the tuned Cross-modal Attention architecture offers the most effective solution for continuous sentiment regression on the CMU-MOSEI dataset.

4.2.3 Performance Comparison Across Unimodal, Bimodal, and Trimodal Fusion Settings

To further understand the contribution of different modality combinations, we compared the tuned tri-modal Cross-modal Attention model against three bi-modal Cross-modal Attention variants: text+audio, text+vision, and audio+vision. We also included the tuned text-only model and Gated Fusion baseline for reference.



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audio+vision vs tri-modal	0.1765	0.2085	< 0.0001	Yes

From the performance comparison, the tri-modal Cross-modal Attention model achieved the best overall result, with the lowest MAE among all settings. Among the three bi-modal combinations, text+vision performed best, followed closely by text+audio, while audio+vision performed substantially worse. This pattern suggests that text remains the

dominant modality in this sentiment regression task, and that both audio and vision provide useful complementary information when combined with text. In contrast, combining only audio and vision without text leads to a large performance drop, indicating that non-textual signals alone are not sufficient to capture sentiment intensity as effectively.

A particularly important result is the comparison between text+vision and the tri-modal model. From a statistical perspective, we do not have sufficient evidence to conclude that the tri-modal model is significantly better than the text+vision combination. In practical terms, this suggests that vision contributes more incremental value than audio once text is already included, and that the performance gain from adding audio on top of text+vision is relatively limited in the current setup.

Taken together, these findings support three main conclusions. First, multimodal fusion is clearly beneficial compared with single-modality modeling, especially when text is combined with at least one additional modality. Second, among the bi-modal settings, text+vision is the strongest and most competitive alternative to the full tri-modal model. Third, while the full tri-modal Cross-modal Attention model remains the best overall configuration, its advantage over text+vision is not statistically strong.

4.3 Model Comparison

The comparison table shows the performance of LSTM-based and Transformer-based models under several representative settings, including the text-only baseline, the cross-modal attention text+vision setting, the cross-modal attention tri-modal setting, and one baseline multimodal fusion model for each backbone.

Table. Comparison of LSTM and Transformer Models Under Representative Settings

Backbone	Setting	MAE	RMSE	Corr
LSTM	text	0.6236	0.8256	0.6672
LSTM	text+vision cross	0.6069	0.8069	0.7017
LSTM	tri-modal cross	0.5975	0.7912	0.7065
LSTM	tri-modal gated	0.6161	0.8165	0.6828
Transformer	text	0.6436	0.8536	0.6408
Transformer	text+vision cross	0.6134	0.8187	0.6783
Transformer	tri-modal cross	0.6056	0.8132	0.6845
Transformer	tri-modal early	0.6215	0.8308	0.6681

The strongest overall performance is achieved by the LSTM-based Cross-modal Attention tri-modal model, which obtains the lowest MAE and RMSE and the highest correlation among all reported results. This suggests that, under the current

experimental setup, the LSTM backbone is more effective at capturing sentiment-related temporal structure in aligned multimodal sequences.

Another important observation is that, for both backbones, the cross-modal attention tri-modal setting performs better than the text-only setting, which confirms that multimodal fusion provides additional useful information beyond the strongest single